

FROZEN DEVELOPMENT WORKFLOW

Mandatory process for every lesson and module

Document	Version	Status	Language
ES-01	1.0	FROZEN	English

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Scope

Applies to every change involving commands, subsystems, IO, telemetry, controls, autonomous, sensors, and utilities.

Step 0 - Backbone Check

Verify package responsibility, dependency direction, control flow, observation flow, RobotContainer role, telemetry rules, and IO rules.

Step 1 - Contract Design

Define interfaces, data snapshots, method signatures, units, ownership, and data sources before implementation.

Step 2 - Architecture Review

Confirm the design preserves the Frozen Backbone, avoids duplicate responsibility, and avoids over-engineering.

Step 3 - Full Implementation

Write code only after review PASS. Every Java change must be delivered as a FULL FILE, never as a diff or ellipsis.

Step 4 - BUILD SUCCESSFUL

Run a build after each clear change. Do not continue while the build is failing.

Step 5 - Simulation / Real Robot Verification

Validate in simulation when applicable, then test on the real robot using safety controls, Glass/Driver Station, and logs.

Stop Rules

- Backbone Check FAIL → stop and redesign.
- Architecture Review FAIL → do not implement.
- Build FAIL → do not advance.
- Verification FAIL → record the cause, fix one change at a time, and retest.

Module Closure

1. Build PASS.
2. Simulation or real robot PASS.
3. Clear Git commit.
4. Record Known Issues and Lessons Learned.
5. Confirm dependencies for the next module.

Revision History

Version	Date	Status	Notes
1.0	2026-07-18	FROZEN	Initial release